

Amazon Review Analytics Using PySpark & NLP

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Overview

This project analyzes Amazon Electronics product reviews using PySpark for distributed data processing and VADER (NLTK) for NLP-based sentiment classification. The analysis covers rating distributions, review trends over time, top-reviewed products, and a comparison of textual sentiment against numerical star ratings. Key findings confirm a strong skew toward positive reviews, a significant growth in review volume in recent years, and high alignment between VADER sentiment labels and star ratings.

Objectives

- Analyze customer rating distributions and trends
- Identify how review activity changes over time
- Identify the most frequently reviewed products
- Perform sentiment analysis on review text
- Compare NLP-derived sentiment against numerical star ratings

Dataset

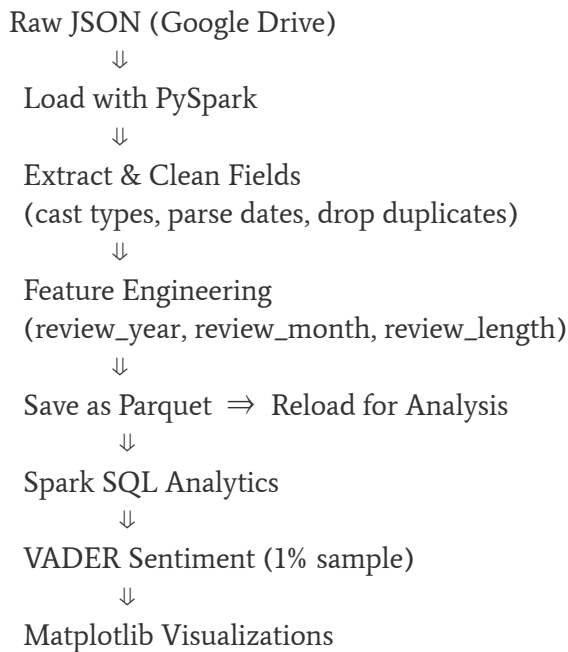
The dataset used in this project is the Amazon Review Dataset provided by Julian McAuley:

<https://nijianmo.github.io/amazon/index.html>

Each record of the Amazon Electronics product reviews includes:

Field	Description
asin	Product ID
reviewerID	Unique reviewer identifier
overall	Star rating (1-5)
reviewText	Full text of the review
reviewTime	Date the review was submitted

Data Pipeline



Key Features

1. Data Engineering

- Processed large-scale JSON data using PySpark
- Converted raw data into structured format
- Removed duplicates and handled missing values
- Created new features such as: `'review_year'`, `'review_month'`, `'review_length'`

2. SQL-Based Analytics

- Rating distribution analysis
- Review trends over time
- Average rating progression
- Identification of top reviewed products

3. Review Behavior Analysis

- Compared review length across rating levels
- Found that lower-rated reviews tend to be longer and more detailed

4. Sentiment Analysis (NLP)

- Applied VADER sentiment analysis to review text
- Classified reviews as Positive, Neutral, or Negative
- Compared sentiment labels against numerical star ratings

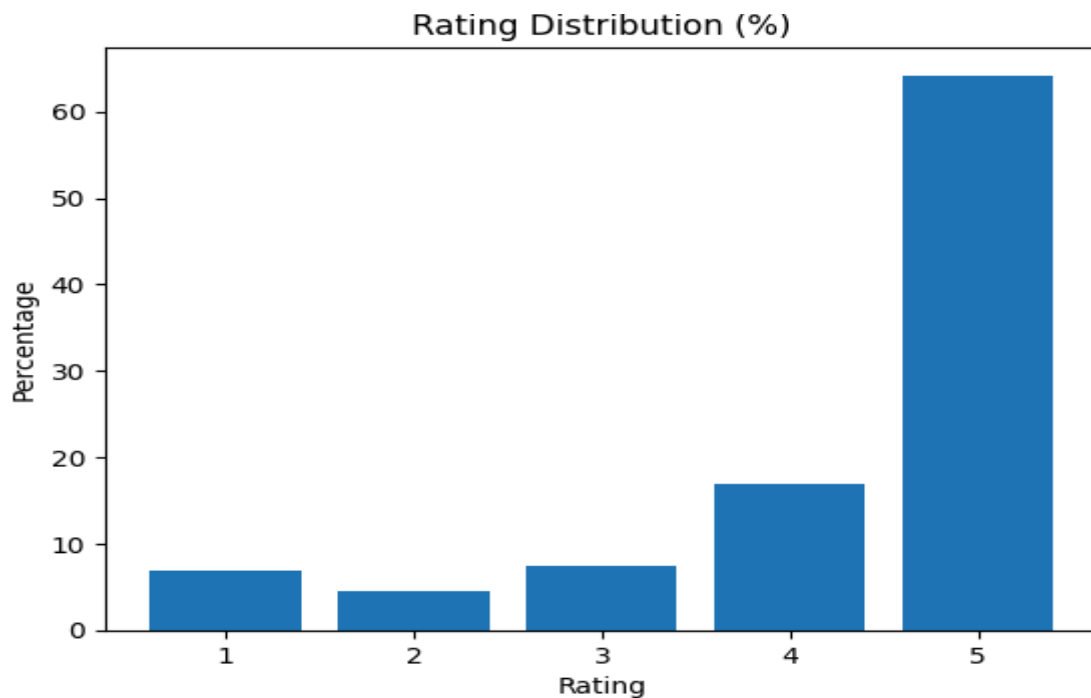
Key Insights

- Review activity increased significantly over time, peaking between 2014–2016
- Average product ratings improved over time, indicating increased customer satisfaction
- Negative reviews tend to be longer than positive ones
- Sentiment and numerical ratings do not always align, highlighting the value of text-based analysis

Analysis & Results

Rating Distribution

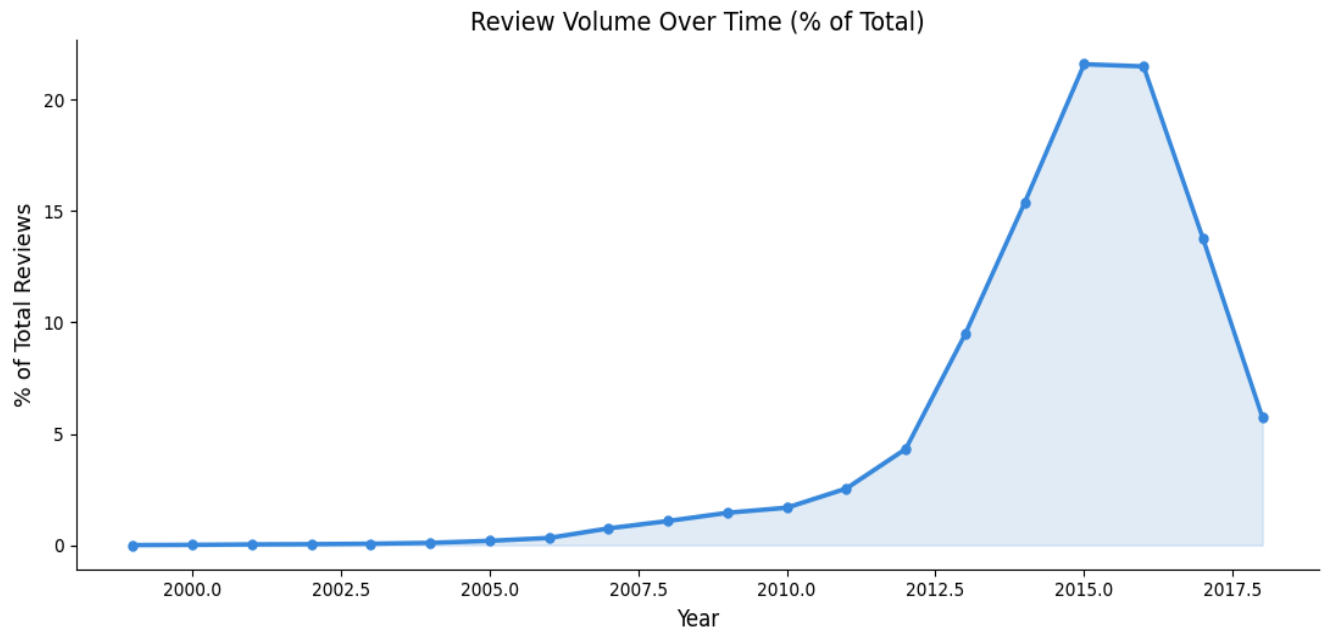
Reviews were grouped by star rating (1–5) and expressed as a percentage of total volume to show the overall distribution of customer satisfaction



The majority of reviews are concentrated at the 5-star rating, indicating a strong positive bias in the dataset. Ratings 1 and 2 account for a much smaller share, suggesting most customers who leave reviews are satisfied with their purchase.

Review Volume Over Time

Review volume was aggregated by year and expressed as a percentage of total reviews to reveal growth trends in customer engagement over time.



Review activity grew substantially in later years, particularly from 2013, reflecting the broader rise of e-commerce and online review culture.

Top Reviewed Products

The 10 most-reviewed product ASINs were identified alongside their average star ratings to compare popularity with customer satisfaction.

index	asin	review_count	avg_rating
0	B003L1ZYYW	8609	4.79
1	B0019HL8Q8	8017	4.76
2	B0019EHU8G	7641	4.76
3	B00DIF2BO2	6222	4.47
4	B00M55C0NS	6219	4.73
5	B00BWF5U0M	6182	4.5
6	B000BQ7GW8	5984	4.71
7	B00IVPU7AO	5983	4.46
8	B00L0YLRUW	5696	4.06
9	B0043T7FXE	5457	4.48

Sentiment vs Star Rating

VADER sentiment analysis was applied to a 1% sample of reviews, classifying each as positive, neutral, or negative. These labels were then compared against numerical star ratings.



NLP sentiment labels align strongly with star ratings, positive sentiment dominates at ratings 4 and 5, while negative sentiment is most prevalent at ratings 1 and 2. The neutral category is distributed more evenly, likely capturing short or ambiguous review text that VADER cannot classify with confidence.

Technologies Used

Tool	Purpose
Python	Core programming language
PySpark	Large-scale distributed data processing
Spark SQL	Declarative analytics queries
NLTK / VADER	NLP-based sentiment analysis
Pandas	Data manipulation for visualization
Matplotlib	Charts and visualizations
Google Colab	Cloud notebook environment

How To Run

Prerequisites

- A Google account with Google Drive access
- The Amazon Electronics dataset (Electronics_5.json), download from [here](#).

Steps

- Upload the dataset to your Google Drive at the following path: [MyDrive/DATA_ANALYTICS_PROJECT/data/Electronics_5.json](#)
- Open Amazon_Review_Analytics.ipynb in Google Colab
- Run the first cell to mount your Google Drive — you will be prompted to sign in and grant access
- Run all remaining cells in order from top to bottom

The following dependencies are installed automatically inside the notebook:

[pip install pyspark nltk](#)

- The cleaned dataset will be saved automatically as a Parquet file at: [MyDrive/DATA_ANALYTICS_PROJECT/data/clean_reviews.parquet](#)

This speeds up any future reruns, you won't need to reprocess the raw JSON again.

Conclusion

This project demonstrates an end-to-end data analytics pipeline, from raw JSON ingestion and distributed processing with PySpark, through SQL-based aggregation, to NLP-driven sentiment analysis and visual reporting. The results validate that customer-assigned star ratings and NLP-derived sentiment are highly consistent, making either a reliable proxy for measuring satisfaction at scale.

Potential next steps include:

- Building an interactive dashboard using Power BI
- Extending sentiment analysis using machine learning models
- Performing product-level deep analysis
- Detecting fake or biased reviews